Statistical Analysis Report: Income and Sex in the 1994 U.S. Census

Author: Francisco Mateus

Dataset: UCI Adult (Census Income) training extract (32,561 rows, 15 columns)

Project: Conduct a Statistical Analysis Using Python capstone (Adult income analysis)

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Overview

I analyzed the UCI Adult (Census Income) dataset (Becker & Kohavi, 1996) to test whether sex is associated with earning over $50K. The dataset is publicly available at https://archive.ics.uci.edu/dataset/2/adult. 7,841 of its 32,561 records (24.08%) earn over $50K.

This report covers the descriptive statistics, four visual models, and one chi-square test of independence. The result is a clear difference between men and women: men in this data earn over $50K far more often than women, 30.57% against 10.95%. The chi-square test says the gap is far too large to be sampling noise alone, with a p-value of about 1e-332. The effect size, Cramér's V, is 0.2160, a small-to-moderate association.

Dataset Description

As noted in the Overview, the analysis uses the UCI Adult training extract (Becker & Kohavi, 1996). It is a real public extract of the 1994 U.S. Census, hosted by the UCI Machine Learning Repository. Each row is one person. The income column records whether that person earns over $50K per year.

The table has 15 columns and 32,561 rows: 6 are numeric (age, fnlwgt, education-num, capital-gain, capital-loss, and hours-per-week) and 9 are categorical. The test uses the two categorical columns, sex and income. The descriptive statistics follow age and hours-per-week, the two numeric columns most relevant to the outcome.

Three columns carry missing values (workclass 1,836, occupation 1,843, and native-country 583, 4,262 cells in total). I keep the missing values in place, because data cleanup is not the purpose of this project. The sex and income columns are complete: all 32,561 rows carry a recorded value for both. The test below therefore uses every row. The missing cells in the other columns do not affect it.

The load check also flags 24 exact duplicate rows. I keep them, because a census extract can legitimately hold two people with identical recorded attributes. One caution comes up later

in the report: the source is a household survey, and the relationship field lists husband, wife, and own-child entries, so the rows may not be independent.

Methods

Lusa et al. (2024) describe the initial data analysis steps that should come before inference: data cleaning, a review of the missing values, and checks on the distribution and shape of each variable. The argument is written for longitudinal studies, but the steps apply to any dataset. Data cleanup stays out of scope, as the Dataset Description notes. Before the test, two checks make up the screening. The load check reports the column types, the missing values, and the duplicate rows. The descriptive statistics show the shape of age and hours-per-week. Those two checks are the whole data preparation, and they found a right-skewed age and two unbalanced variables (income and sex). The choices below follow from them.

For the question itself I use a chi-square test of independence. The question asks whether sex and income are associated rather than how much one predicts the other. Both variables are categorical, which is exactly what the test expects. The p-values come from a standard approximation that is reliable only when the expected counts are large (Agresti, 2007). The expected counts here are far above that level, as the Results show.

The null hypothesis (H0) is that sex and income are independent. The alternative (H1) is that they are associated. I fixed the significance level at alpha = 0.05 before the test.

A chi-square test is non-directional: it can say whether an association exists, but not which way it runs. I read the direction from the observed counts and rates instead. A second assumption is that every row is a separate observation. That assumption is not fully true here: the source is a household survey, and records can share a household. The p-value is therefore approximate (Agresti, 2007). The Limitations section returns to this concern.

SciPy's chi2_contingency applies the Yates correction by default to a two-by-two table. I report the uncorrected Pearson statistic and note the corrected value in the Results. The effect size uses the uncorrected statistic. As an effect size I report Cramér's V, a 0-to-1 measure of the

strength of an association (Ben-Shachar et al., 2023). The p-value alone does not show that strength, and it is easy to misread as the probability that the null hypothesis is true. I therefore report the p-value and the effect size together (Greenland et al., 2016).

All computations used Python 3.12.14 with pandas 3.0.6, NumPy 2.5.3, and SciPy 1.18.1 (Virtanen et al., 2020), and the figures used Matplotlib 3.11.2 and Seaborn 0.13.2. The notebook runs top to bottom from a clean kernel, writes the four figures itself, and prints every number the report quotes. requirements.txt pins the exact package versions. The notebook's Reproducibility section walks through the whole rebuild with uv, step by step.

Results

The descriptive statistics show the shape and balance of the four relevant variables:

- age is right-skewed: the mean is 38.5816, the median is 37.0, the standard deviation is 13.6404, the interquartile range is 28.0 to 48.0, and the skewness is 0.5587.
- hours-per-week concentrates at a 40-hour week: the mean is 40.4375, the median is 40.0, the standard deviation is 12.3474, the interquartile range is 40.0 to 45.0, and the skewness is 0.2276.
- income is unbalanced, with 7,841 people (24.08%) over $50K.
- sex is uneven, with 21,790 men and 10,771 women.

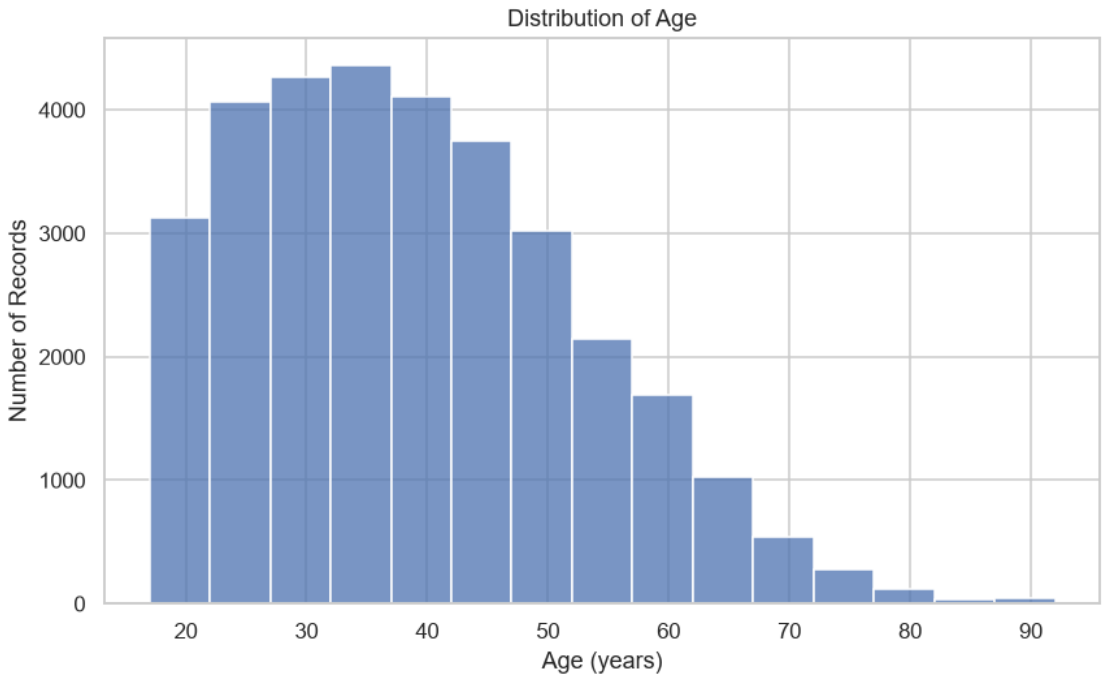


Figure 1. Distribution of age, showing the right skew.

The curve in Figure 1 falls the way the skewness number predicts, but the last populated bin needs a separate explanation. The notebook's supporting counts show it holds 47 records, with 43 at the 90 cap, 3 at 88, and 1 at 87. The data appear top-coded at 90 (respondents beyond the cap recorded as 90), so that bump reflects the cap rather than a natural peak of older respondents.

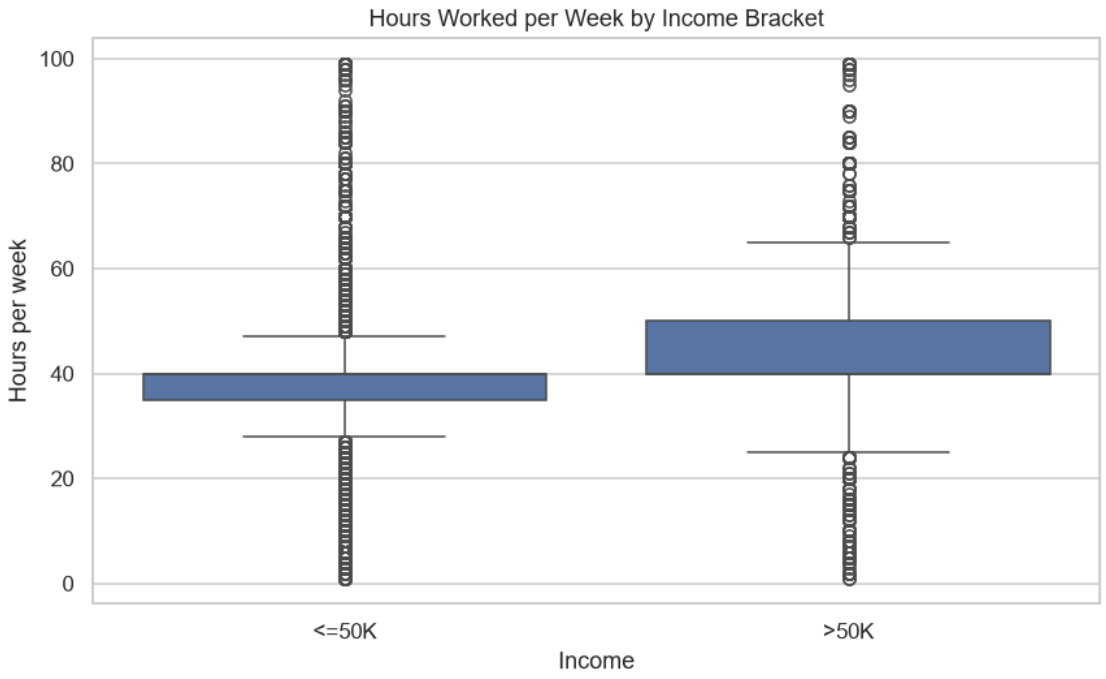


Figure 2. Hours worked per week by income bracket.

Figure 2 compares how weekly hours are spread between the two income groups.

- **Median:** 40 hours in both groups.
- **Middle range:** 40 to 50 hours in the over-$50K group, compared with 35 to 40 hours in the at-or-below-$50K group.
- **Past 40 hours:** 49.2% in the over-$50K group, compared with 23.2% in the at-or-below-$50K group.

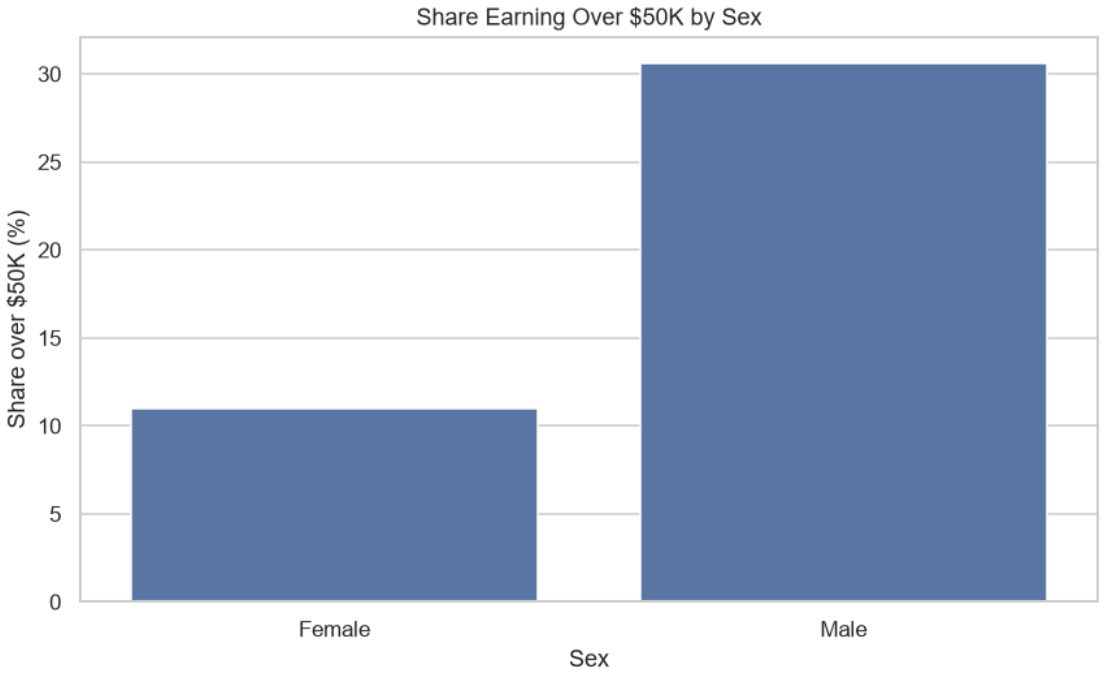


Figure 3. Share earning over $50K by sex.

Figure 3 shows the income rate by sex, the clearest single view of the question. 30.57% of men earn over $50K against 10.95% of women, a gap of 19.63 percentage points.

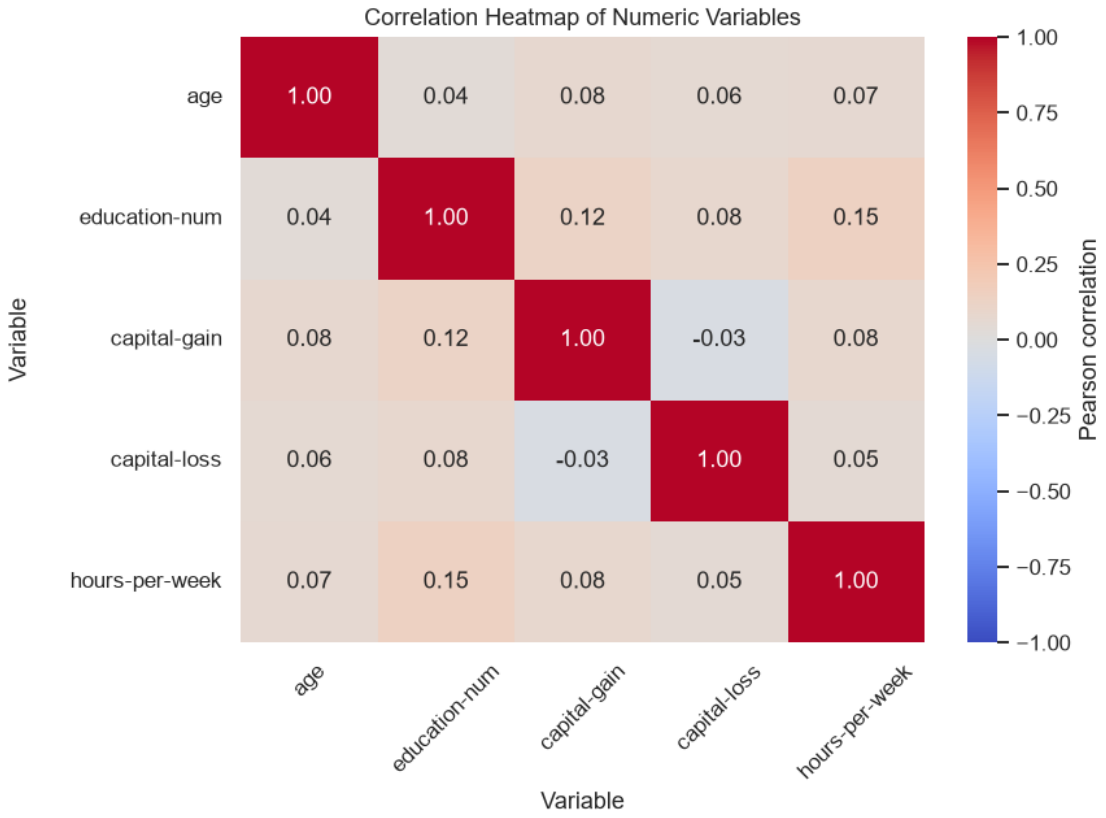


Figure 4. Correlation heatmap of the numeric variables.

Figure 4 shows the correlations among the plotted numeric variables.

- education-num and hours-per-week are the strongest link: 0.1481.
- income is categorical and outside the matrix, so this view does not directly answer the income question.
- fnlwgt is omitted because it is a census sampling weight, not a measured attribute.
- capital-gain and capital-loss are mostly zeros (91.67% and 95.33%), so their correlations are not informative.

The chi-square test rejects independence.

- Pearson statistic: 1518.8868; Yates-corrected: 1517.8134; degrees of freedom: 1.
- p-value: about 1e-332, far below 1e-300.

- Smallest expected count: 2593.76, so the approximation is safe.
- Effect size: Cramér's V 0.2160; V squared 0.0466 (4.7%).

The contingency table is:

Sex	At or below $50K	Over $50K
Female	`9,592`	`1,179`
Male	`15,128`	`6,662`

The over-$50K rate is 19.63 percentage points higher for men than for women. Its 95% confidence interval spans [18.78, 20.48] (Agresti, 2007). The interval and the p-value both assume independent observations, so possible household clustering makes both approximate.

Interpretation for a Non-Technical Audience

In this dataset, men were much more likely than women to earn over $50K. About three in ten men were above the threshold, against about one in ten women, a gap of about 19.63 percentage points.

A gap of 19.63 points in a sample of 32,561 would almost never appear by chance if sex and income were unrelated, and the chi-square test confirms the gap (p about 1e-332). The effect size shows how strong the association is: 0.2160 is a small-to-moderate value on the 0-to-1 scale, and sex alone explains only about 4.7% of the variation in who is above the line. Sex is part of the difference, not all of it.

The result is an association, not proof that sex caused the difference in earnings, and it applies to the 1994 United States data. The Limitations section lists the other differences between the groups and the other caveats.

Limitations and Potential Bias

This is an observational comparison, so the result is an association, not proof of sex causing the difference.

The 1994 U.S. Census extract and its undocumented sampling design limit how far the

result generalizes. The $50K threshold is nominal and not inflation-adjusted, and sex is recorded as a binary field.

Missing values in three columns and the possibility of household-linked rows make the observed rates, p-value, and confidence interval approximate. A follow-up model using education-num, occupation, and hours-per-week, accounting for those issues, would be a useful next step.

References

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